

TraceMail AI

Email Forensic Investigation Report

Case ID: TM-190012DC

Generated: 2026-09-06T01:39:11.669193

1. Executive Summary

Severity	CRITICAL
Risk Score	100/100
Classification	Business Email Compromise / Financial Phishing

Assessment: The email presents a high-confidence set of suspicious indicators and should be treated as a potentially malicious communication. Further investigation and containment are recommended.

2. Email Evidence

Filename	
Sender	finance@paypa1-security.com
Recipient	employee@company.com
Reply-To	urgent-payment@protonmail.com
Subject	URGENT: Invoice Payment Required Today
Date	Sat, 05 Sep 2026 10:30:00 +0530
Message-ID	<123456789@example.com>

3. Email Authentication Findings

Authentication Results	mail.company.com; spf=fail; dkim=fail; dmarc=fail
Received-SPF	fail
DKIM Signature	False

4. Header & Relay Forensics

Total Relay Hops: 1

Earliest Observed Public IP: 8.8.8.8

Observed Public IPs:

• 8.8.8.8

The earliest observed public IP represents observed source infrastructure and should not automatically be treated as the attacker's physical location.

5. IP Intelligence

IP	Country	City	ISP	Risk
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8.8.8.8	United States	Mountain View	Google	LOW
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6. URL & Domain Intelligence

Extracted URLs: 1

- <http://paypal1-security.com/invoice/84721>

Extracted Domains: paypal1-security.com

7. Threat DNA

Feature	Value
spf	FAIL
dkim	FAIL
dmARC	FAIL
url	True
url_count	1
financial	True
urgency	True
reply_to_mismatch	True
attachment	False
hop_count	1
typosquatting	True

DNA Fingerprint:

SPF-FAIL|DKIM-FAIL|DMARC-FAIL|URL-1|FIN-1|URG-1|RPLY-1|ATT-0|HOP-1|TYPO-1

8. Identified Risk Factors

- {'type': 'authentication', 'severity': 'high', 'message': 'SPF authentication failed'}
- {'type': 'authentication', 'severity': 'medium', 'message': 'DKIM authentication is missing or failed'}
- {'type': 'authentication', 'severity': 'high', 'message': 'DMARC authentication failed'}
- {'type': 'header', 'severity': 'high', 'message': 'Reply-To domain differs from sender domain'}
- {'type': 'content', 'severity': 'medium', 'message': 'Urgency language detected', 'matches': ['urgent', 'immediately', 'action required', 'today']}
- {'type': 'content', 'severity': 'high', 'message': 'Financial/payment-related intent detected', 'matches': ['payment', 'invoice', 'account']}
- {'type': 'url', 'severity': 'medium', 'message': '1 URL(s) detected in email'}

9. Evidence Integrity

Hash Algorithm: SHA-256

Evidence SHA-256:

59f1dc7bcd5926080987f1d49f631903072c96b47ff0b1fb6b92c4ae3c40ee61

SHA-256 hash uniquely identifies the analyzed email evidence and supports tamper-evident integrity verification.

10. Investigator Conclusion

The email presents a high-confidence set of suspicious indicators and should be treated as a potentially malicious communication. Further investigation and containment are recommended.

Generated by TraceMail AI — Email Threat Detection & Forensic Intelligence Platform